

Indian Statistical Institute
Second Semester Examinations: 2022-23

Course Name: M. Math, 1st year
Subject Name : Topology II
Maximum Marks: 40, Duration: Two hours
Date: 24.02.2025, 2:30 PM - 4:30 PM

- Answer as many questions as you can.
- Maximum marks is 40.
- You may use any results proved in class. Any other results (including those in homework problem sets) require proof.

1. Let X be a CW complex such that $X = A \cup B$ with A, B subcomplexes, where each of A, B and $A \cap B$ are contractible. Prove that X is contractible. 8

2. Show that the complement of a finite set of points in $\mathbb{R}^n$ is simply connected if $n \geq 3$. 10

3. Let n be the set $\{0, 1, \dots, n\}$. For $0 \leq i \leq n$ Consider the abstract simplicial complex $(V, \mathcal{F}_i)$ defined by

$$V = n, \mathcal{F}_i = \mathcal{P}(n) \setminus \{n, n \setminus \{i\}\}.$$

Show that the geometric realization $|(V, \mathcal{F}_i)|$ is contractible. 10

4. Consider maps $X \xrightarrow{f} Y \xrightarrow{g} Z$ with X, Y and Z path connected and locally path connected. Suppose that both $g \circ f$ and f are covering spaces. Prove that g is a covering space. 7

5. Consider the action of the group $C_p = \langle a \mid a^p = 1 \rangle$ (where p is an odd prime) on $X = S^{2n+1} \times S^2 = \{(z_0, \dots, z_n, z) \mid z_i \in \mathbb{C}, \sum |z_i|^2 = 1, z \in \hat{\mathbb{C}} = \mathbb{C} \cup \infty\}$ given by the formula (ζ is a fixed p^{th} root of unity)

$$a((z_0, z_1, \dots, z_n), z) = ((\zeta z_0, \dots, \zeta z_n), \zeta z) \text{ (where } \zeta \cdot \infty = \infty).$$

Let $q : X \rightarrow X/C_p$ be the usual quotient map. Fix the basepoint $*$ of X and hence of X/C_p as the point $((1, 0, \dots, 0), 0)$.

(a) Compute $\pi_1(X/C_p, *)$. 4

(b) Let Y be a space with maps fitting in the diagram below

$$\begin{array}{ccccc} & & q & & \\ & \nearrow & & \searrow & \\ X & \xrightarrow{f} & Y & \xrightarrow{g} & X/C_p \end{array}$$

Suppose that f is a covering space. Prove that either $Y \cong X$ or $Y \cong X/C_p$. 5

$$\partial_2(\sigma_1) = \sigma_1 - \sigma_2 = g_* f_* \sigma_1 = g_* (f_* \sigma_1)$$